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Investigation of the development of the Robotic Dexterous Hands

Plans based on Tetheria Gen1 Open Source Hand

Design and Optimization of Flexible-Rigid Composite Dexterous Hand
For Safe Grasping and Mechanical Protection

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1. Foundation & Baseline Establishment

Assembly and test Tetheria Hand

Gathering operation data and identifying limitations

2. Optimisation

Design multifunctional pads and safety mechanisms

Comparative testing and analysis



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3D printing all components

Purchasing

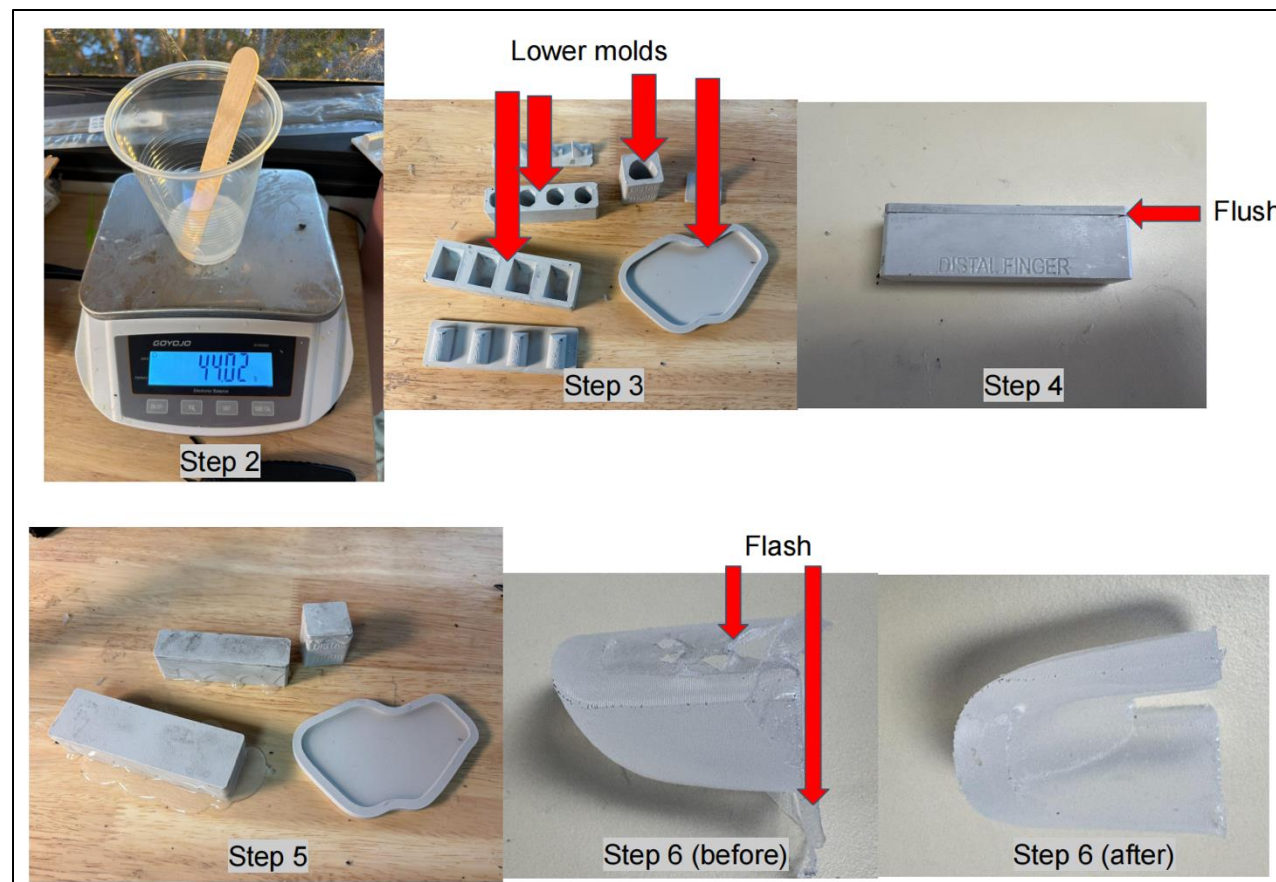
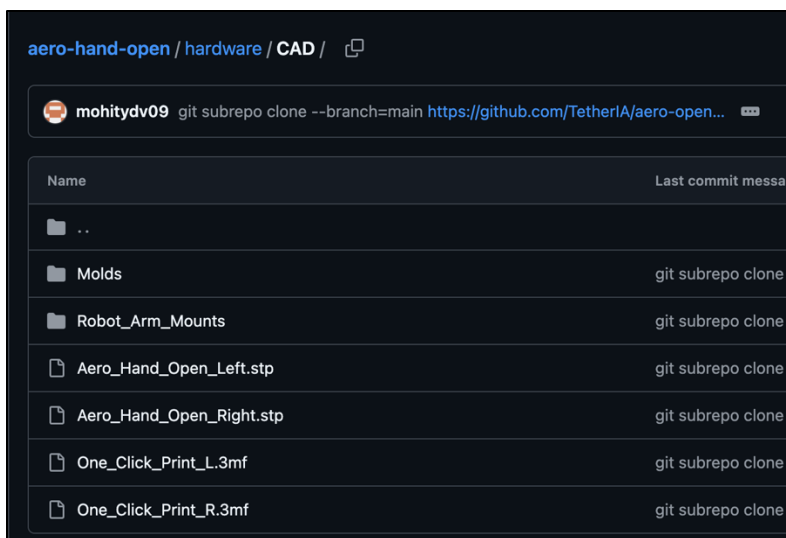
Assembly

Hardware Setup

First-Time-Setup

Python SDK

	P/N	Name	Material	Qty per Hand
3D Printed		1 Finger Base	PLA	4
		2 Finger MCP	PLA	4
		3 Finger Proximal	PLA	4
		4 Finger Medial	PLA	4
		5 Finger Distal	PLA	4
		6 Cable Spool	PLA	6
		7 Thumb Abduction Upper Servo Disk	PLA	1
		8 Thumb Abduction Lower Servo Disk	PLA	1
		9 Thumb Abduction Linkage Bar	PLA	1
		10 Thumb CMC Base	PLA	1
		11 Thumb MCP	PLA	1
		12 Thumb Proximal	PLA	1
		13 Thumb Distal	PLA	1
		14 Servo Frame	PLA	1
		15 Palm Front Frame	PLA	1
		16 Palm Rear Frame	PLA	1
		17 Silicone molds Top	PLA	1
		18 Silicone molds Bottom	PLA	1



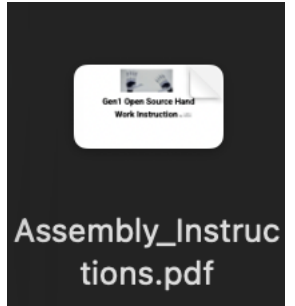


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Purchased Components	23	Finger Distal/Medial and Thumb Proximal/Distal Return Spring (PIP)	Music Wire	5
	24	Finger Proximal/MCP Return Spring (MP)	Music Wire	4
	25	Thumb CMC/MCP Return Spring (CMC)	Music Wire	1
	26	Finger Pull Cable	Kevlar/Vectran	3
	27	Pinky Pull Cable	Kevlar/Vectran	1
	28	Finger Coupling Cable	Kevlar/Vectran	4
	29	Thumb CMC Flex Cable	Kevlar/Vectran	1
	30	Thumb Pull Cable	Kevlar/Vectran	1
	31	Thumb Coupling Cable	Kevlar/Vectran	1
	32	2x5x2.5 bearing	Steel	23
	33	1x10 Pin	Steel	16
	34	obsolete (2x10 Pin)	Steel	0
	35	2x14 Pin	Steel	12
	36	2x20 Pin	Steel	2
	37	UNDERSIZED 2x10 Pin	Steel	19
	38	UNDERSIZED 2x20 Pin	Steel	1
	39	UNDERSIZED 2x30 Pin	Steel	1
	40	M2x6 Flanged Torx Self Tapping Screw	Steel	25
	41	M2x10 Countersunk Torx Self Tapping Screw	Steel	12
	42	M3x10 Rounded Torx Self Tapping Screw	Steel	9
	43	M3x5.7 Heat set insert	Brass	4
	44	Feetech HLS3606M Servo Motor	Comes with purchase of servo from Feetech	7
	45	Servo Motor Mount Plate		
	46	Servo Motor Spline Screw M2.3x5		
	47	Servo Motor Plate Screw M2x4		
	48	Connection Board and Cable		
	49	TTLinker Mini UART Signal Conversion Board	PCB	1
	50	ESP32-S3	PCB	1

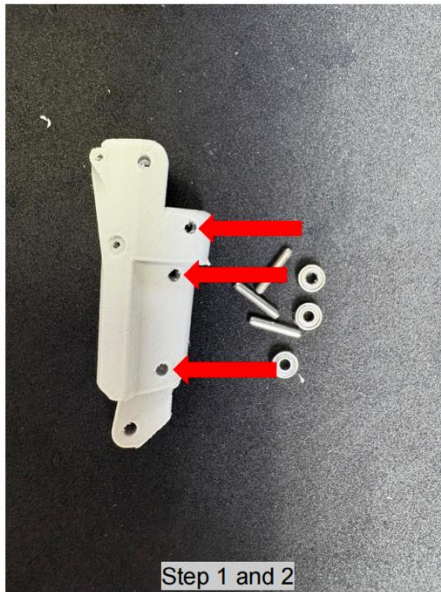


Drill

1. Drill d
Note



Bearings

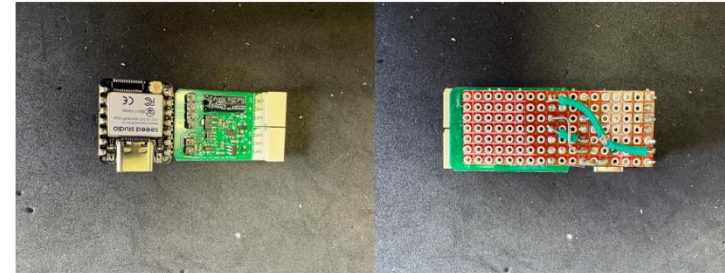


Control Board

(not needed if kit purchased)

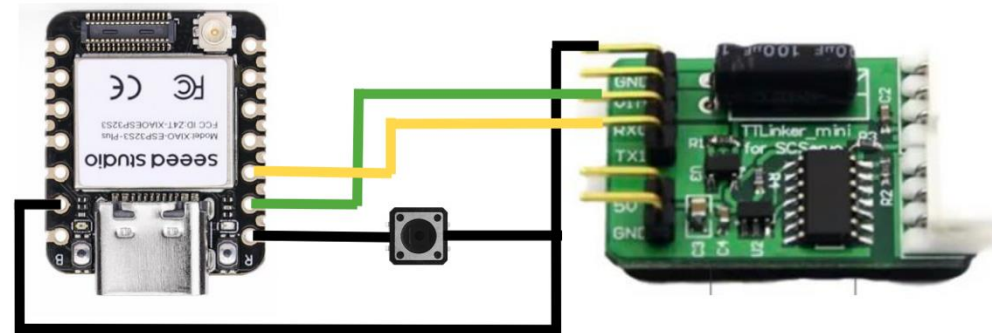
2. Build a control board using the electrical schematic

Board can be built on a breadboard (no soldering required), custom PCB (coming soon) or protoboard



Example protoboard

GREEN - RX
YELLOW - TX
BLACK - GROUND





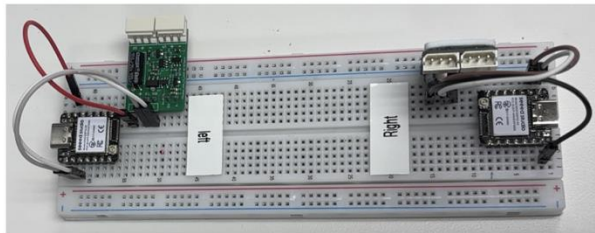
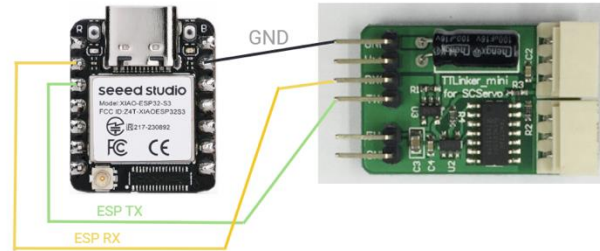
Next Step: Solder Pin Headers to your ESP32 S3



Once ESP is Soldered, Try to use Multimeter to check if any connections are not shorted Together.

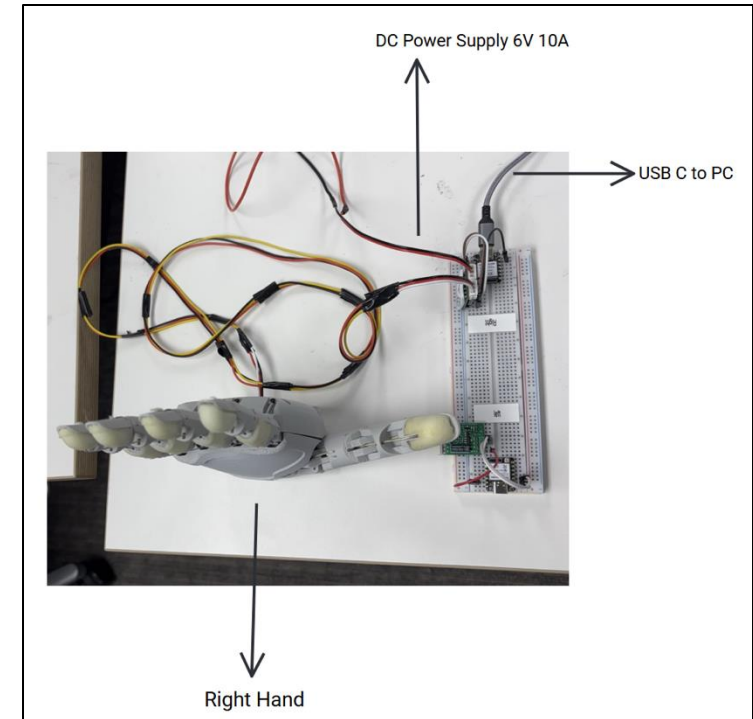
Take out 3 Pin to Pin headers and the TTLLinker Mini Servo Board. Do the Connections on ProtoBoard

ESP TX ↔ TX
ESP RX ↔ RX
GND ↔ GND



Once Setup is Done, It should look like this.

Note: This setup is do to control 2 hands , Left to Control Left hand.
Right Indicates the same setup to control right hand.
It contains the same TTLLinker mini board, Just at different angles.



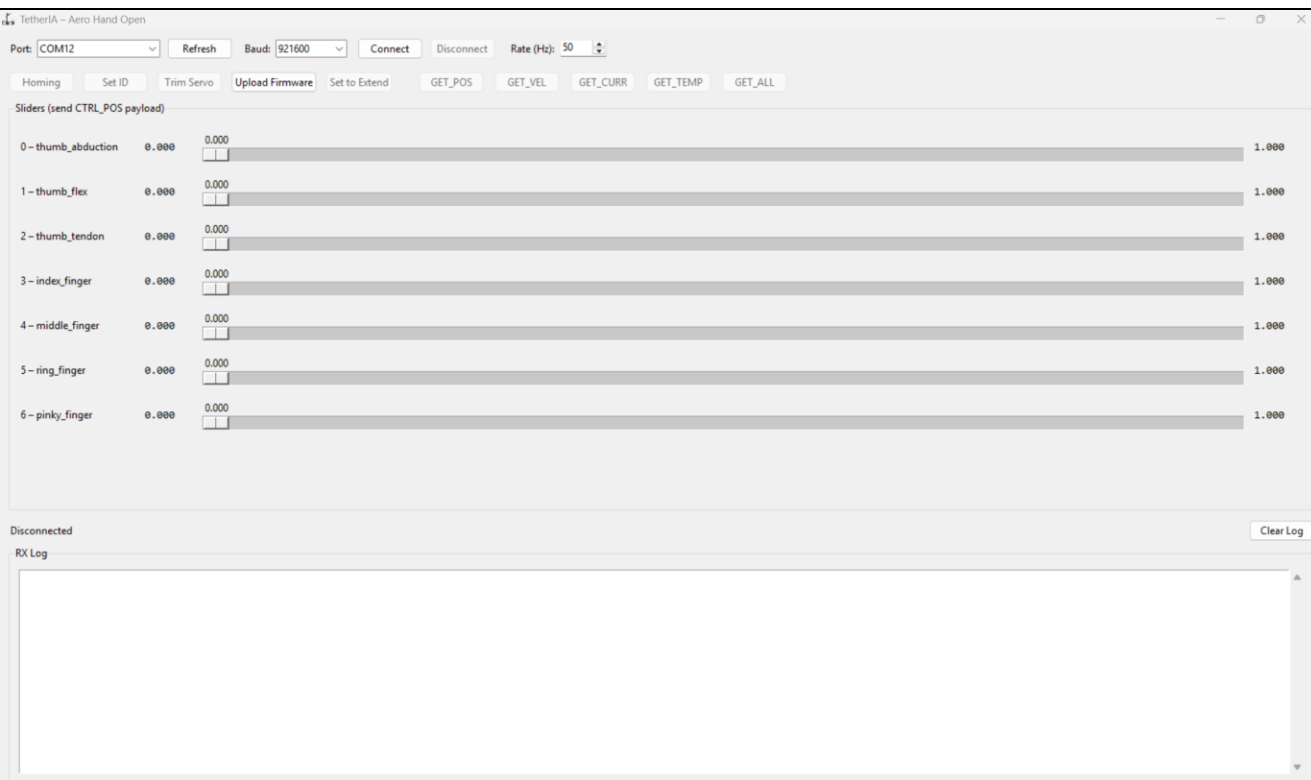


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GUI — Aero Hand Open Control App



Controlling the Hand

- Actuation Position Control
- Actuation Torque Control

Receiving Data from the Hand

- Get Actuator Currents
- Get Actuator Temperatures
- Get Actuator Speeds

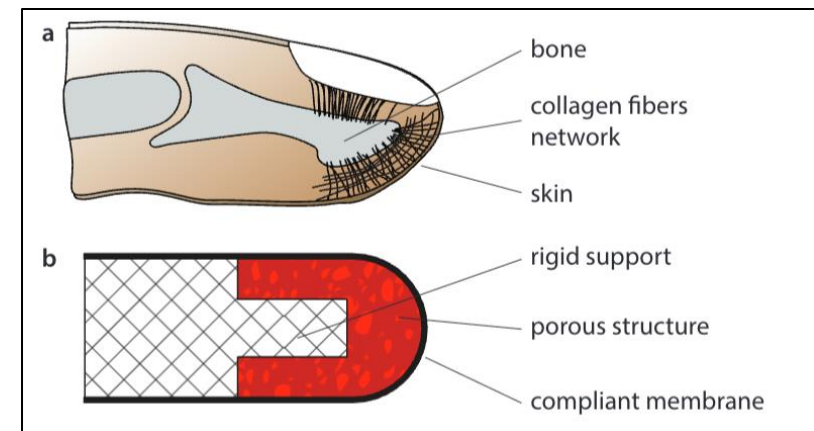
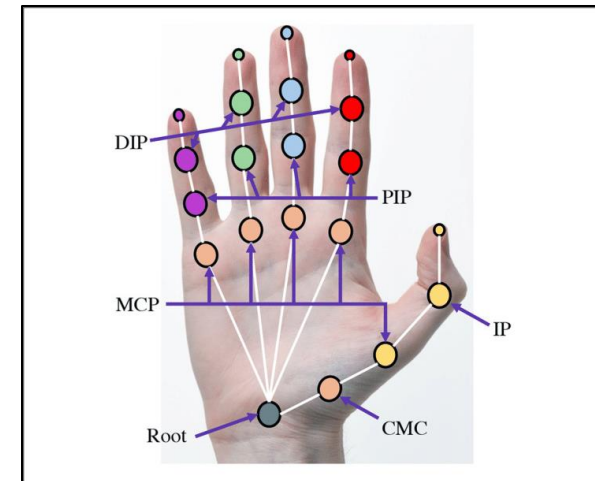


Outcomes:

- Having better understanding through the assembly of Dexterous Hand
- Gathering Data

Objectives:

- Design and manufacture multi-functional finger pads and palm pads that can improve gripping stability
- Design mechanical "fuses" at critical joints (MCPs & CMC) to prevent servo damage from accidental collisions or jamming





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Appendix

<https://github.com/TetherIA/aero-hand-open?tab=readme-ov-file#resources>

<https://docs.tetheria.ai/docs/intro>